Gauthier HUMEAU

Final-Year Engineering Internship – Embedded Systems / Systems Engineering
6 months starting February 2027 | Aerospace, space, defense | Open to international mobility
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SUMMARY

Final-year engineering student (Master's level) at IPSA, Systems track. Hands-on projects in real-time embedded software (ESP32, FreeRTOS), control systems (Amesim, Simulink) and robotics (ROS2). Student pilot (PPL in progress) with field experience in aircraft maintenance. Fluent English (C2), American international section.

EDUCATION

Master's-level Engineering Degree (Diplôme d'Ingénieur) – Final year 2022 – 2027
IPSA, Aerospace Engineering School – Systems track, major in Space, Launchers and Satellites *Paris, France*
Relevant coursework: real-time systems, microcontrollers, digital electronics (VHDL, FPGA), control engineering, signal processing, avionics and data buses, A320 electrical systems.

Exchange Semester – Computer Science: data science, big data, cryptography Sep 2025 – Feb 2026
Hanoi University of Science and Technology (HUST) *Hanoi, Vietnam*

French Baccalaureate, International Option, with honors – American Section 2019 – 2022
Lycée Raoul Follereau – majors in Mathematics and Physics-Chemistry *Belfort, France*

TECHNICAL PROJECTS

Embedded Free-Fall Detector – ESP32, FreeRTOS, C/C++
- Data acquisition from an MPU6050 IMU (I2C) and an INMP441 microphone (I2S) on an ESP32 Feather; real-time fall detection.
- Multitask architecture under FreeRTOS; push notifications to a smartphone over Wi-Fi.

Landing Gear Control System – Amesim, MATLAB/Simulink
- Cylinder sizing (25 kN under 206 bar), closed-loop PID control: 500 mm stroke, 9 mm overshoot.

Autonomous Multi-Robot Exploration – ROS2, Gazebo, Python, Linux
- Three autonomous robots covering a room; A* path planning implemented in Python, simulated in Gazebo and RViz.

A320 Home Cockpit (personal project) – Arduino, MobiFlight, 3D printing
- Built a nose-wheel steering tiller on an Arduino Pro Micro with existing firmware, integrated into the flight simulator via MobiFlight; 3D-printed parts (Flashforge).

Satellite Image Classification with Deep Learning – Python, TensorFlow/Keras
- MobileNetV2 CNN made robust to simulated sensor and atmospheric noise, benchmarked against a Random Forest.

Real-Time Fraud Detection – Spark, Kafka, MongoDB, Docker
- Lambda architecture for batch and streaming: approximately 2,000 transactions/min processed, with over 400 "smurfing" alerts raised by Spark; monitoring via Grafana.

PROFESSIONAL EXPERIENCE

Business Engineer – Internship Jun 2026 – Sep 2026
Cleeven, engineering and technology consulting firm *Basel, Switzerland*
- Recruited engineers and technicians across industries (about 12 candidates interviewed per week); industrial prospecting.

Aircraft Maintenance Intern Aug 2024
Pays de Montbéliard Airfield *Montbéliard, France*
- Preventive and corrective maintenance on single-engine aircraft; release to service, logbooks, airworthiness directives.

SKILLS

Embedded and electronics: C/C++ (Arduino, ESP32), FreeRTOS, I2C, I2S, VHDL (DE10-Lite FPGA, Quartus).
Simulation and software: MATLAB/Simulink, Amesim, Python (TensorFlow), ROS2, Linux, Git, Docker, CATIA.
Languages: French (native); English, fluent (LILATE C2, TOEIC 920/990); Spanish (basic, A2).

LEADERSHIP AND INTERESTS

Aviation: French Aeronautics Initiation Certificate (BIA), Private Pilot License (PPL) in progress, flight simulation.
Secretary of the IPSA Student Union – one of 4 executive board members (40 members, about 1,000 students).
Sports: half marathon, basketball, judo. **Other:** French citizen; driver's license.